

Department of Leather Engineering
Khulna University of Engineering & Technology
B. Sc. Engineering 3rd Year 2nd Term Examination-2021
ME 3221
Manufacturing Process

Time: 3.0 Hours

Full Marks: 210

- N.B. i) Answer any **THREE** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.
iii) Assume reasonable data if any missing.

SECTION-A

- | | | |
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| 1(a) | What is casting? Write down the advantages and disadvantages of sand casting. | 06 |
| 1(b) | Compare and contrast between removable and disposable patterns. | 10 |
| 1(c) | Explain the properties of a good core. | 10 |
| 1(d) | Describe pressure casting process with neat sketches. | 09 |
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| 2(a) | What are the factors that affect the permeability of a sand mold? | 04 |
| 2(b) | Explain the reciprocating type continuous casting process with necessary figure. | 16 |
| 2(c) | What is solid phase welding? What are the advantages of welding over other joints? | 09 |
| 2(d) | Draw various types of welding joints. | 06 |
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| 3(a) | Explain projection welding process with advantages, limitations and applications. | 18 |
| 3(b) | Explain the characteristics of various flames produced during oxyacetylene gas welding. | 10 |
| 3(c) | What are the functions of a gas welding torch? | 07 |
| | | |
| 4(a) | Why metal forming process is necessary? Classify metal forming process. | 08 |
| 4(b) | Write short note on: (i) Piercing, (ii) Forging and (iii) Shearing. | 12 |
| 4(c) | How recrystallization takes place during Hot Rolling? Draw various rolling arrangements and mention their significance. | 15 |

SECTION-B

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|------|---|----|
| 5(a) | What is manufacturing process? Explain how the knowledge of manufacturing process can be used in leather and footwear industries. | 10 |
| 5(b) | What is a machine tool? How do you classify manufacturing process based on surface finishing operation? Describe. | 07 |
| 5(c) | Distinguish between orthogonal and oblique cutting. | 08 |
| 5(d) | What is machining? Explain the principle of metal cutting. | 10 |
| | | |
| 6(a) | Derive the expression for shear angle and shear strain in orthogonal cutting model. | 10 |
| 6(b) | What is tool life? For a given set of conditions, how would you arrive at the best cutting speed for maximum tool life? Explain. | 10 |

6(c)	From the machining performance view point, which type of chip is preferred? Explain your answer with suitable justification. Also, show the conditions which favor such a chip formation.	10
6(d)	What are the functions and properties of cutting fluid?	05
7(a)	Using neat sketches, describe the various operations that can be carried on lathe machine.	10
7(b)	Explain the basic working principle of a Lathe machine.	12
7(c)	Explain the quick return mechanism with neat sketch. What are the differences between shaper and planer?	10
7(d)	Write short notes on counter boring.	03
8(a)	Explain the different elements of a twist drill with neat sketch.	12
8(b)	What is indexing of milling? What are the types of indexing? Describe any one of them.	10
8(c)	What is universal milling machine? Show the nomenclature of a milling cutter.	08
8(d)	How do you specify a grinding wheel?	05

Department of Leather Engineering
Khulna University of Engineering & Technology
 B. Sc. Engineering 3rd Year 2nd Term Examination-2021
LE 3211
Footwear Engineering II

Time: 3.0 Hours

Full Marks: 210

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 ii) Figures in the right margin indicate full marks.
 iii) Assume reasonable data if any missing.

SECTION-A

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|------|---|----|
| 1(a) | What is meant by shoe last? How does a shoe last reflect the distinctive anatomy of the human foot? | 05 |
| 1(b) | Explain the role of toe spring in a shoe last. | 08 |
| 1(c) | Provide a visual representation of different girths of shoe last with explanation. | 10 |
| 1(d) | Write short notes on the following shoe lasts:
(i) Three-piece last (ii) Sports shoe last and (iii) Last for veldtschoen construction | 12 |
| 2(a) | Assume that a 37-size shoe last has last length of 246 mm and joint girth of 209 mm. Create a table with the values for last length and joint girth using both arithmetic and geometric grading, for Paris point sizes 34-42. | 08 |
| 2(b) | How to carry out co-ordination grading of shoe last? Mention the pros and cons of co-ordination grading. | 10 |
| 2(c) | State the reason behind providing the master patterns to the clicker before cutting shoe materials. | 07 |
| 2(d) | Define leather sorting and grading. Why leather sorting and grading is required before clicking operation? | 10 |
| 3(a) | What are the quality standards needed to maintain before cutting vamp and toe cap for shoe making? | 08 |
| 3(b) | What is meant by the instruction sheet? How can the competence of a clicking operator be measured with the assistance of instruction sheet? | 07 |
| 3(c) | Classify the shoe material cutting methods. State the advantages of travelling head clicking machine over swing arm press machine. | 06 |
| 3(d) | Write short note on die-less cutting system. | 07 |
| 3(e) | Define closing. Why is size-marking required before closing operation? | 07 |
| 4(a) | "Making of dies for shoe upper cutting with a number of features can increase the productivity of shoe industry"- Explain it. | 07 |
| 4(b) | How to carry out stitch marking on shoe upper using a loose die machine? | 06 |
| 4(c) | Briefly explanation the methods of gimping operation. What are the points to watch while carrying out gimping process? | 08 |
| 4(d) | Classify seam. Distinguish between Barca and Bordo seams. | 08 |
| 4(e) | Define closing. Mention different types of seam used in moccasin shoe construction. | 06 |

SECTION-B

- 5(a) Explain lasting principles in terms of material process and design for successful lasting operation. 10
- 5(b) Describe the following lasting methods with figure: 10
(i) Rib lasting and (ii) Force lasting
- 5(c) Sketch and explain the drafting directions of hand lasting process. 10
- 5(d) What are the basic points to look for in seat and side lasting operation? 05
- 6(a) What are the points to be remembered before lasting process 05
- 6(b) What do you know about Veldschoen shoe construction? What are the basic characteristics of this construction? 10
- 6(c) Write down the temperature and dwell time required in back part moulding in case of both upper softening and setting shape. What are the problems faced during stiffener attachment? 10
- 6(d) Explain the hand lasting method including its pulling direction. 10
- 7(a) Write down the sequence of operations for Goodyear Welted shoe construction. 10
- 7(b) Differentiate Veldschoen shoe construction with Goodyear welted shoe construction in terms of lasting process. 10
- 7(c) Define vulcanizing. Why raw rubber is vulcanized before using it as a soling material? 08
- 7(d) Explain the following operations for Veldschoen shoe construction: 07
(i) Rough Rounding (ii) Edge pairing and (iii) Edge scouring
- 8(a) What are functions of roughing and scouring in lasting process? 08
- 8(b) Differentiate between direct and indirect shoe constructions. 07
- 8(c) Point out the problems, possible cause and remedy for attaching toe puffs in lasting process. 10
- 8(d) Explain upper preparation process for: (i) Coated leather and (ii) PU coated leather. 10

Department of Leather Engineering
Khulna University of Engineering & Technology
 B. Sc. Engineering 3rd Year 2nd Term Examination-2021
LE 3209
Leather Products Engineering II

Time: 3.0 Hours

Full Marks: 210

- N.B. i) Answer any **THREE** questions from each section in separate scripts.
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SECTION-A

- 1(a) What is meant by fabrication technology? State the major stages of fabrication technology for leather goods. 08
- 1(b) Briefly describe the sequence of screen printing on leather. Mention the instruction and acceptance criteria for creasing and cementing operations of leather goods construction. 10
- 1(c) Clarify the terms- embossing, engraving and debossing. 06
- 1(d) Define seam efficiency. A leather jacket has three stitches per centimeter along the seam line. A specimen was cut according to ISO 3376 with a dimension of 110 × 25 mm from the jacket having 0.9 mm thickness. The tensile load reading from a tensile tester (model no. STD 172) was 200 N for the specimen and leather tensile strength was given 18.0 N/mm². Compute the percentage of seam efficiency and explain the result. 11
- 2(a) Edge construction is very important to distinguish leather products manufacturing – why? 10
- 2(b) Cite the nesting principle for leather goods. List out the conventional methods of material consumption. 05
- 2(c) Show the 90° clockwise nesting technique of front panel for a ladies bag. Derive the relation; $f_c = \left(\frac{W_2}{S^2} - \frac{1}{A} \right) S + 1$ for a specific condition; where the symbols have their usual meanings. 08
- 2(d) PICARD Bangladesh Ltd. is a world class leather wallet manufacturer. During the last calendar year, the cost incurred for a wallet fabrication is as follows: direct materials 300 Tk; direct labor 650 Tk; manufacturing overhead 320 Tk; transportation overhead 170 Tk. The labor force involved in the operation process was 12. Each labor is paid a wage of 325 Tk. In addition to the salaries, the factory spends 6,500 Tk. per year on forms, postage, etc. The feasible purchase order will be 1500. Calculate conversion cost, total cost of production for the wallet, and also the difference in activity rate. 12
- 3(a) State the thread consumption parameters. 06
- 3(b) What is thread consumption? Write down the factors influencing thread consumption. 08
- 3(c) How do you calculate stitch length? 04
- 3(d) Prove that $C_{301} = S (2L + 2T) \pm e$; where symbols have their usual meanings. 06

- 3(e) A thread is used to stitch a ladies' purse made out of vegan leather with a thickness of 0.9 mm material thickness. Given that S.P.C. value is 3.0. Calculate thread consumption for lock stitch and also by applying the above data show that; $C_{401} > C_{301}$ 11
- 4(a) Why poromerics are called third generation synthetic material regarding structure and performance? 08
- 4(b) Cite the different parameters of basic needle system. 04
- 4(c) Discuss the indirect system of thread numbering. 10
- 4(d) Remark on color code of needle. Given blade diameter of two needles are 0.024 in. and 0.035 in. respectively. Determine the corresponding European and American needle sizes. 13

SECTION-B

- 5(a) Explain the stages of new product development. 10
- 5(b) Write down the classification of a sewing needle according to the needle points. 10
- 5(c) Mention some sewing problems occurred during sewing process with their possible remedies. 10
- 5(d) Write short note on feed dog. 05
- 6(a) What are the required qualities needed for a lining material in leather products manufacturing? 10
- 6(b) Differentiate among interlining, interfacing and underlining. 10
- 6(c) Why mercerization is done on fabric and what are the specific qualities improved through this process? 10
- 6(d) Make the differentiation between warp and weft. 05
- 7(a) Define feed. Describe the feeding mechanism of sewing needle. 10
- 7(b) Explain the selection criteria of lining for a leather jacket. 07
- 7(c) Briefly describe the basic structure of a zipper. 08
- 7(d) Describe the factors that affect fusing process. 10
- 8(a) State the origin, properties and uses of lining materials. 10
- 8(b) Write down the mechanism of lock stitch sewing process. 10
- 8(c) Write down the advantages of using fusible interlining. 10
- 8(d) What are the differences between woven and non-woven fabric? 05

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Department of Leather Engineering
Khulna University of Engineering & Technology
 B. Sc. Engineering 3rd Year 2nd Term Examination-2021
LE 3219
Leather Microbiology

Time: 3.0 Hours

Full Marks: 210

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SECTION-A

- | | |
|--|----|
| 1(a) What is bacteriology in microbiology? | 03 |
| 1(b) Write down the chemical composition and function of the following bacterial particular structure: i) Cell wall, ii) Ribosome and iii) Fimbria. | 09 |
| 1(c) In a petri dish, 1.2×10^4 , no. of bacteria were found. After 10 days, the numbers of bacteria were increased and the numbers were 1.8×10^4 . Find out the number of generations and generation time of the bacteria. | 12 |
| 1(d) What is the percentage law of bacterial death? Explain the growth curve of bacteria. | 11 |
| 2(a) Describe the reproduction and spore formation in bacteria. | 12 |
| 2(b) Define culture media. Classify the types of culture media with examples and their uses. | 08 |
| 2(c) Explain the differences between pour plate and streak plate methods. | 05 |
| 2(d) What is active immunity? Mycobacterium cannot be stained by Gram staining procedure. Justify the statement. | 10 |
| 3(a) Assume that you have to isolate and characterize any salt tolerant colony from tannery wastewater. Now write the sequential steps of the procedure. | 12 |
| 3(b) Why do gram-positive and gram-negative bacteria show different color in staining operation? | 07 |
| 3(c) Write down the role of bacteria in the leather industry. | 06 |
| 3(d) Write down the name of five defects caused by bacteria, fungi, molds and also mention their effect on the leather. | 10 |
| 4(a) Mention the physical, chemical and mechanical microbial control methods. | 10 |
| 4(b) How do antimicrobial agents work? Explain. | 12 |
| 4(c) Write short notes on: (i) B-cell, (ii) Sterilization, (iii) Pasteurization and (iv) Antibody. | 08 |
| 4(d) What is the anaerobic culture method? Write down the chemical composition of synthetic media. | 05 |

SECTION-B

- | | |
|---|----|
| 5(a) What are hyphae? Write down the morphology of fungal hyphae. | 10 |
| 5(b) What is symbiosis in mycology? Give an example that explains the symbiotic relationships of fungi. | 08 |

5(c)	What are spores? Write on the fungal mitospores.	10
5(d)	Distinguish between Plasmogamy and Karyogamy.	07
6(a)	State briefly the components of media for industrial fermentation.	08
6(b)	What are fermenters? State the ideal features of a fermenter.	10
6(c)	Distinguish between submerged and surface techniques of fermentation.	08
6(d)	Write down the working principles of Air lift bioreactor with its advantages and limitations.	09
7(a)	Define the terms: enzyme activity, specific activity, turnover number and enzyme assay.	08
7(b)	State the covalent modification method of enzyme activity regulation.	08
7(c)	Write down the primary screening methods for the identification of Catalase and Amylase enzymes producing microorganisms.	10
7(d)	What is immobilization of enzymes? Give the advantages and limitations of immobilized enzymes.	09
8(a)	Show the basic differences between lactic acid and alcoholic fermentation.	10
8(b)	Write about the requirements of ideal environment for the growth and multiplication of fungi.	10
8(c)	State the enzyme inhibition process using Line weaver-Burk plot.	10
8(d)	Write the importance of microbiology in leather making.	05

Department of Leather Engineering
Khulna University of Engineering & Technology
 B. Sc. Engineering 3rd Year 2nd Term Examination-2021
ME 3219
Fluid Mechanics & Machineries

Time: 3.0 Hours

Full Marks: 210

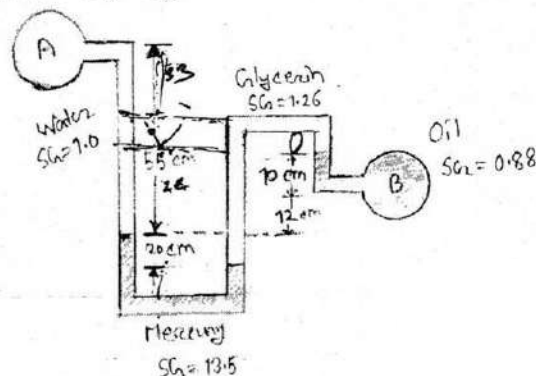
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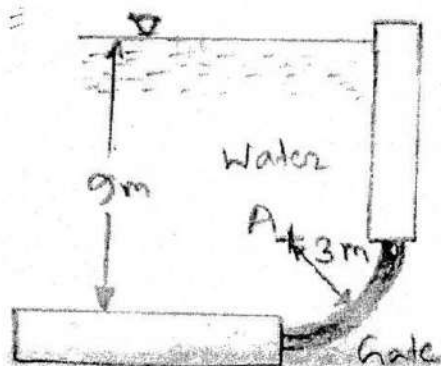
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SECTION-A

- 1(a) What is fluid? Define following terms: 10
 (i) Viscosity (ii) Newtonian fluid and non-Newtonian fluid and (iii) Compressibility
- 1(b) Describe and classify various types of fluids and fluid motions. 15
- 1(c) Describe surface tension and capillary effect of surface tension of a fluid. 10
- 2(a) Derive the fundamental equation of fluid in rest. 15
- 2(b) What are the devices typically used to measure fluid pressure? How pressure is measured using a piezometer tube? 08
- 2(c) The pressure difference between an oil pipe and water pipe is measured by a double-fluid manometer as shown. For the given fluid heights and specific gravities, calculate the pressure difference, $4P = P_B - P_A$ 12



- 3(a) Distinguish absolute, gauge and vacuum pressure. Also indicate their relative positions on a single chart. 10
- 3(b) Describe the necessary criteria of immersed and floating bodies. 10
- 3(c) A 4 m long curved gate is located in the side of a reservoir containing water. Determine the magnitude of the horizontal and vertical components of the force of the water on the gate. 15



- 4(a) State and explain Reynolds transport theorem. Apply Reynolds transport theorem to mass conservation and deduce the continuity equation. 15
- 4(b) Gasoline at 20°C is pumped through a smooth 12 cm diameter pipe of 10 km long, at a flow rate of 75 m³/h. The inlet is fed by a pump at an absolute pressure of 24 atm. The exit is at standard atmospheric pressure and is 150 m higher. Estimate the frictional head loss. 20

SECTION-B

- 5(a) Define static, stagnation, dynamic and total pressure. 10
- 5(b) Derive the Euler equation of fluid flow considering a non-turbulent, non-viscous flow. 15
- 5(c) Derive Bernoulli's equation from Euler equation of fluid flow. 10
- 6(a) Classify various types of hydraulic pumps. 10
- 6(b) Describe the working principle of a reciprocating pump. 10
- 6(c) A double-acting reciprocating pump running at 40 rpm is discharging 1.0 m^3 of water per minute. The pump has a stroke of 400 mm. The diameter of the piston is 200 mm. The delivery and suction head are 20 m and 5 m, respectively. Find the slip of the pump and power required to drive the pump. 15
- 7(a) Define the following terms: 09
(i) Flow separation (ii) Specific speed and (iii) Manometric efficiency
- 7(b) Why centrifugal pump is termed as roto-dynamic pump? Is it advantageous over reciprocating pump? – Justify answer. 12
- 7(c) Find the power required to drive a centrifugal pump which delivers 50 liters of water per second to a height of 20 m through a 150 mm diameter and 100 m long pipeline. The overall efficiency of pump is 70% and Darcy's $f = 0.06$ for the pipeline. Assume inlet losses in suction pipe equal to 0.30 m. 14
- 8(a) What is cavitation? How is it formed? 08
- 8(b) What are the troubles that can occur in a centrifugal pump? Also name out the way of remedies. 10
- 8(c) Distinguish among fan, blower and compressor. 06
- 8(d) How compressor is classified? Shortly describe screw compressor and centrifugal compressor. 11